ASSIGNMENT 1

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Subject: Cryptology

Question 1

Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

1. The security feature/service required, such as confidentiality, integrity, authentication, authorization, or availability.
2. The security mechanism used to provide the identified security feature.
3. The security protocol or technology used to implement the mechanism.
4. Briefly explain how the mechanism/protocol provides the identified security feature.

Answer

Here are five common applications and online services that use security features to protect users and their information.

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
WhatsApp	Confidentiality	End-to-End Encryption	Signal Protocol	WhatsApp encrypts messages before sending them. The message can only be read by the sender and the person receiving it. This helps keep private conversations safe from others.
Gmail	Authentication	Multi-Factor Authentication	OTP / TOTP / FIDO2	Gmail can ask for an extra verification step after entering the password. For example, the user may need to enter an OTP or approve the login on their phone. This makes it harder for someone else to access the account.
Online Banking	Confidentiality and Authentication	Encryption and MFA	HTTPS / TLS and OTP	Online banking uses HTTPS and TLS to protect information while it travels between the user's device and the bank's server. OTP is also used to confirm the user's identity, especially during important transactions.
Google Drive	Authorization	Access Control	OAuth 2.0	Google Drive allows users to decide who can view or edit their files. OAuth 2.0 is used to give applications limited access without sharing the user's password. This helps prevent unauthorized access to files.
Online Shopping	Confidentiality and Integrity	Secure Communication	HTTPS / TLS	When we shop online, details such as login information and payment information are sent through an encrypted connection. HTTPS and TLS help protect this information from being read or changed while it is being transmitted.

Conclusion

Security is important in almost every online service we use today. Different applications use different security methods depending on what they need to protect. Encryption keeps information private, authentication checks who the user is, and authorization controls what the user is allowed to access. These methods help make online communication and transactions safer.

Question 2

For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Answer

I looked at five online services that people commonly use in daily life. Each service has different security controls depending on the type of information it handles. Some controls stop an attack before it happens, while others help identify a problem or restore the service after an attack.

Application	Security Classification	Security Control / Mechanism	Protocol / Technology	How it Helps
WhatsApp	Confidentiality, Integrity	End-to-End Encryption	Signal Protocol	Messages are encrypted before they are sent from the user's phone. Only the intended person can read them. This makes it difficult for someone who intercepts the communication to understand the message.
Gmail	Confidentiality, Integrity	Spam filtering, malware detection and account security	TLS, SPF, DKIM, DMARC	Gmail checks emails for spam, harmful attachments and suspicious links. TLS protects email data while it is being transferred. SPF, DKIM and DMARC help check whether an email is really from the sender it claims to be from.
Online Banking	Confidentiality, Integrity, Availability	Encryption, transaction verification and monitoring	HTTPS/TLS, OTP, MFA	TLS protects banking information while it is being sent. OTP or another verification method is used for important transactions. Banks also monitor unusual transactions to identify possible fraud.
Google Drive	Confidentiality, Availability	Access control, file permissions and backups	OAuth 2.0, TLS	Users can decide who can view, comment on or edit their files. OAuth 2.0 provides controlled access without directly sharing the user's password. Backups also help users recover their files if something goes wrong.

Application	Security Classification	Security Control / Mechanism	Protocol / Technology	How it Helps
Online Shopping	Confidentiality, Integrity, Availability	Secure payment, fraud detection and server protection	HTTPS/TLS, PCI DSS	HTTPS protects information such as passwords and payment details during transmission. Payment systems check transactions for unusual activity. Reliable server systems also help keep the shopping service available to users.

Preventing, Detecting and Recovering from Attacks

Security controls do not all have the same purpose. Some are used before an attack happens, while others help identify a problem or recover from it.

Preventing:

Encryption, MFA, access permissions and secure connections make it harder for attackers to access accounts or steal information.

Detecting:

Spam filters, malware scanners, login alerts and unusual-transaction monitoring can help identify suspicious activity.

Recovering:

Backups, account recovery systems and reliable servers help restore data or services after an attack or system failure.

Conclusion

From these examples, it can be seen that online services normally use more than one security control. For example, an online banking system can use encryption to protect information, OTP to verify the user and transaction monitoring to notice suspicious activity. Using several controls together provides better protection than depending on only one security method.